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Unsupervised Learning



Supervised Learning

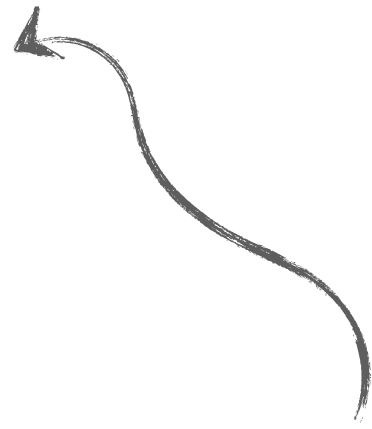


Reinforcement Learning

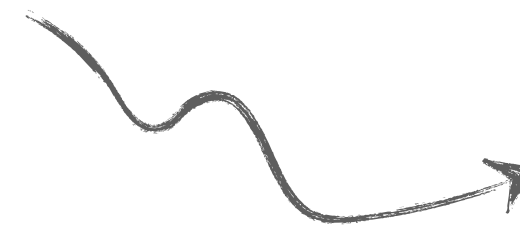


TAXONOMY OF FINANCIAL TASKS

Example- A computer first clusters patient data to find groups with similar health patterns using unsupervised learning. Then, using supervised learning, it predicts the likelihood of a specific illness within each group based on labeled cases of that illness in the past.



Example- A computer is trained on past data to suggest fitness routines for heart patients. As patients follow these suggestions, the system refines recommendations based on their health progress (reinforcement learning).



Example- A public health system employs unsupervised learning to group regions with similar vaccination rates. In regions with lower rates, it utilizes diverse outreach tactics and applies reinforcement learning to identify the most effective strategies, adjusting its methods accordingly.

3

4



SUPERVISED LEARNING

Task The Machine wants to predict whether an individual will develop Type 2

diabetes has a direct link to heart disease.



:BodyMassIndex(

Experience—Supervise what we have and as set of individuals with the following

:Fasting blood sugar level



[Label]: Diabetes Status (0 for No, 1 for Yes)

x

1

x2

BNI





, where:

is a vector representing the health indicators of the

Formal Presentation–We represent our data as a collection of pairs:

individual development disabilities.

represents whether the

$$\{ (x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)}) \}$$

$$x^{(i)} = [x_1^{(i)}, x_2^{(i)}]$$

ith

$y(i)$

i^oth

Performance—Once the model is trained, its performance is assessed on new,

unseen data to gauge how well it predicts the onset of diabetes.

35